

PROJECT ROLES & RESOURCE ALLOCATION

Developer: [Left Blank] | OptiCrop Project Documentation

Repository: <https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine>

1. Project Developer Profile

The OptiCrop project was developed entirely as an individual project. The developer executed all roles across the software development lifecycle:

Development Role	Responsibilities Executed	Key Deliverables Completed
Product Lead	Requirements elicitation, planning, and documentation.	SRs, empathy mapping, and user journey maps.
ML Engineer	Data preprocessing, algorithm selection, and training.	Trained model pipelines and evaluation plots.
Backend Developer	Flask server controller logic, route validation, database logic.	api.py, routes, predictions.csv, health check routes.
UI/UX Designer	Glassmorphic dashboard style guidelines, responsive layout.	index.html, result.html, dashboard.html templates.

Member A - Lead Presenter
Member A - Lead Presenter
Delivers introduction, problem framing, closing roadmap & Q&A.

Member B - ML Presenter
Member B - ML Presenter
Explains dataset, model selection, accuracy metrics, confusion matrix.

Member C - App Presenter
Member C - App Presenter
Performs live web-app demo: form input -> crop result -> dashboard.

Member D - QA Presenter
Member D - QA Presenter
Runs pytest live, shows CSV log, explains test coverage.

Member D - QA Presenter
Runs pytest live, shows CSV log, explains test coverage.